

Beyond Sentiment: The Influence of Game Pricing on Player Feedback in Steam Reviews

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Abstract—The vast amount of user-generated content in the gaming industry presents a challenge for developers seeking to understand player feedback. This study investigates the relationship between a game's pricing and the focus of user discourse in reviews. We address the question of whether a game's price influences player sentiment and the topics they discuss. After using BERT [1]-based sentiment analysis on a large-scale dataset of 730,945 Steam [2] reviews across 743 games to classify the user sentiment, we employed a manual, keyword-based topic analysis to classify review content and analyzed these findings across different price segments. The study reveals that price explains only a negligible fraction ($\approx 0.2\%$) of sentiment variation, suggesting that pricing alone does not determine overall positivity or negativity in reviews, but it significantly shapes player expectations. Topic analysis reveals clear shifts in the aspects players emphasize across price tiers: low-priced games elicit detailed reviews focusing on value and story; mid-priced games concentrate on gameplay and immersion; and high-priced games generate fewer reviews and narrower discussions. These findings highlight that while price does not predict sentiment, it serves as a signal that frames how players evaluate and articulate their experiences.

Keywords—Sentiment Analysis, Topic Analysis, Game Price Analysis, BERT, Steam Reviews, OLS.

I. INTRODUCTION

The gaming industry is evolving every day. The video game industry has grown into a global entertainment powerhouse, with its success driven in large part by a vibrant ecosystem of user-generated content, particularly in the form of online reviews. Platforms such as Steam [2] provide a direct line of communication between players and developers, offering a real-time source of feedback. However, the vast number of unstructured reviews makes it very challenging to find out the player demands on various aspects of a game.

Prior research on game reviews has largely focused on sentiment classification, recommendation systems, or price prediction. However, less attention has been paid to the role of pricing as a factor that may shape how players evaluate and discuss games. Specifically, it remains unclear whether a game's price influences (1) the sentiment expressed in reviews and (2) the aspects of gameplay and design that players choose to emphasize. We hypothesize that while a direct price-to-sentiment correlation may be weak, price fundamentally influences player expectations, which is reflected in the topics and themes of their reviews. Addressing this gap is important for both researchers and developers, as pricing is a fundamental design and marketing decision.

This study aims to investigate the relationship between a video game's price, the review sentiment and content. We ask two central questions :

(RQ1): Does a game's price affect the user's sentiment expressed in user reviews? (RQ2): Does a game's price act as a signal that sets player expectations, and if so, how is this reflected in the topics and themes of player feedback?

Our research moves beyond simple sentiment classification to provide a nuanced analysis of player priorities.

To address these questions, we conducted a comprehensive analysis of over 730,945 reviews from 743 paid games on the Steam platform. We applied BERT[1]-based sentiment analysis to assess whether price correlates with sentiment and employed a manual, keyword-based topic analysis to classify review content into categories such as gameplay, story, graphics, immersion, and value for money.

Our findings show that price explains only a negligible fraction ($\approx 0.2\%$) of sentiment variation, indicating that pricing does not determine whether reviews are positive or negative. However, topic analysis reveals systematic shifts across price tiers: low-priced games elicit detailed reviews emphasizing story and value, mid-priced games center on gameplay and immersion, and high-priced games generate fewer reviews and narrower discussions. These results suggest that while price does not predict sentiment, it frames the dimensions of evaluation that players prioritize. This contributes to ongoing work in game analytics by clarifying the nuanced role of pricing in shaping—not sentiment itself—but the structure of player discourse.

A. Objective

- Try to find out any correlation between price and user sentiment.
- Find out what user talk about the most in different price segments.

II. LITERATURE REVIEW

Steam reviews provide a rich source of user-generated data that reflects both player satisfaction and broader market trends. Existing research has leveraged these reviews for diverse purposes, but the potential link between game pricing and user sentiment remains largely unexplored.

One line of work has focused on authenticity and spam detection. For example, Bian et al. [3] (2021) examined the reliability of Steam reviews and reported that approximately 15% of 33,450 reviews could be classified as spam.

Another stream has emphasized aspect-based sentiment analysis. Tran and Huy [4] (2024) developed a machine learning framework that clustered reviews into gameplay aspects and applied an RNN-based classifier, while Busurkina [5] et al. (2021) used Structural Topic Models (STM) to evaluate different dimensions of player experience.

In addition, topic modeling has been widely applied to large-scale review datasets. Yu[6] et al. (2021) employed Latent Dirichlet Allocation (LDA) on 1.2 million esports reviews, while Guzsvinecz [7] (2025) analyzed 418,483 reviews of Soulslike titles using LDA to uncover recurring themes. Li[8] et al. (2021) also used topic modeling in a data-driven framework to assess game playability.

Together, these studies demonstrate the richness of insights available from Steam reviews. However, they have largely ignored how game pricing interacts with user sentiment and review content. To our knowledge, this dimension has not been systematically explored, leaving a research gap that the present study addresses.

III. METHODOLOGY

As stated in the related works section, there is no prior research to our knowledge that correlates price with user sentiment and topics of discussion. To bridge that gap, we collected game IDs and corresponding user reviews of 762 games. Because of public APIs availability, Steam was chosen as the main source of our data collection.

A. Data Collection

To collect the game ids and user reviews, the SteamSpy API[9] was used and stored in CSV files for further processing. Using a simple python script, paid games were filtered out and divided into 3 sections(Fig. 1),

1. High-priced games: >40 USD
2. Mid-priced games: 20-40 USD
3. Low-priced games 1-20 USD.

Using the filtered game IDs, we collected up to 1000 multilingual user reviews per game. To filter out low-quality or spam reviews, only reviews with at least 5 words were retained, ensuring that each review contained meaningful content.

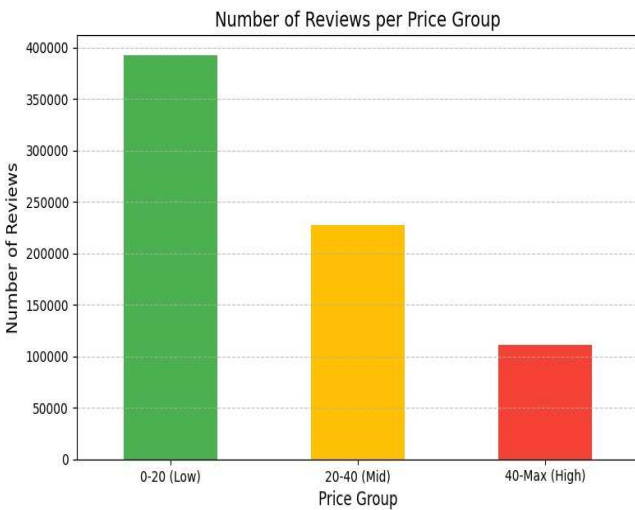


Fig. 1. Number of reviews in each price segment

The reviews were stored in separate CSV files according to their price category, which were later merged for analysis. The dataset includes columns such as game_ID, price, review text, numeric sentiment, and playtime. Additional metadata such as release date, developer and publisher were collected but not analyzed in this study.

This demonstrates that lower priced games have more user interaction than higher priced games.

B. Data Preprocessing

For manual keyword-based topic selection and sentiment analysis, the raw reviews were preprocessed using the NLTK [10], [11] libraries to remove stopwords, numbers and punctuations.

C. Sentiment Analysis

To analyze the correlation between pricing and player sentiment, we used a BERT-based sentiment analysis model to the merged dataset (Fig. 2). Reviews were classified into 3 separate categories,

- 5★, 4★: Positive
- 3★: Neutral
- 2★, 1★: Negative

These sentiment labels formed the basis for subsequent analyses.

D. Evaluation of Multilingual BERT-based Sentiment Analysis

To ensure the reliability of our sentiment predictions, we employed a Multilingual BERT (mBERT[12]) model (Fig. 3), which is pre-trained on 104 languages and capable of handling diverse text inputs.

The model was evaluated using standard performance metrics including accuracy, precision, recall, and F1-score on a validation set of labeled reviews. Where

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

$$Precision = \frac{TP}{TP+FP}$$

$$Recall = \frac{TP}{TP+FN}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Here,

TP = True Positives

TN = True Negative

FP = False Positives

FN = False Negatives.

High scores across these metrics indicate that the model effectively captures the underlying sentiment of user reviews, even in the presence of mixed-language content, providing a robust foundation for analyzing correlations between game price and player sentiment.

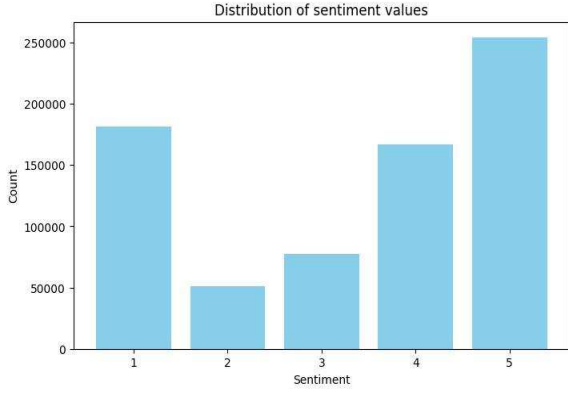


Fig. 2. Distribution of review sentiment across all collected reviews.

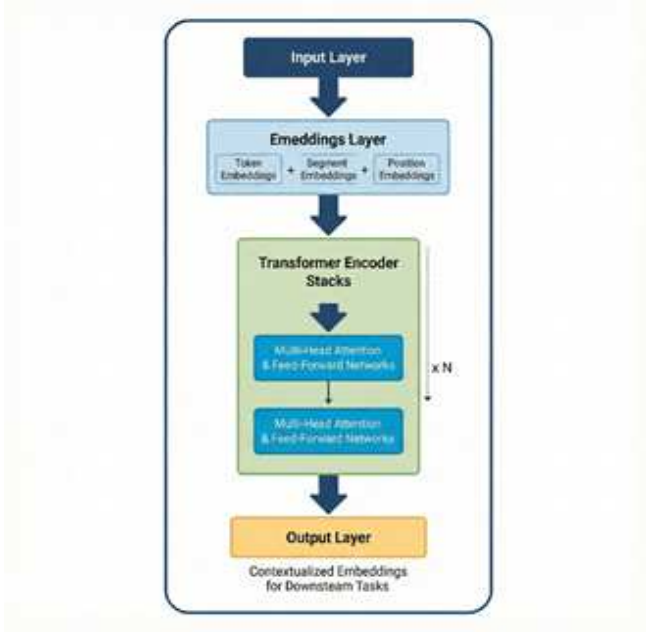


Fig. 3. mBERT architecture

E. Workflow Diagram

The procedures used in this article are shown in “Fig. 4”.

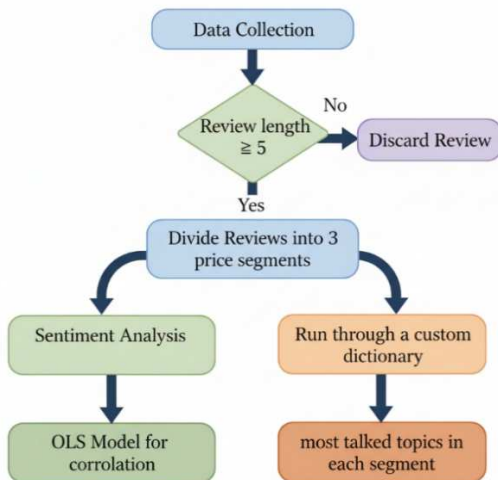


Fig. 4. Workflow Diagram

At first, user reviews along with other metadatas was collected from Steam using the games id list found via SteamSpy api. To ensure quality, reviews containing fewer than five words are excluded from further analysis. The collected reviews were then divided into 3 price segments. Each segment is subjected to two parallel analyses: (i) sentiment analysis, which is followed by the application of an OLS regression model to investigate the relationship between sentiment and price, and (ii) processing through a custom dictionary to determine the topics that are most frequently discussed within each segment.

F. Dataset

We collected a dataset of 730,945 user reviews from 743 paid Steam games, with approximately 1,000 reviews per game. Each review record included the review text, author playtime (in hours), and game price, along with additional metadata such as votes, timestamps, and game details from the SteamSpy API. For the purposes of our analysis, only the review text, and price were utilized, while other metadata (e.g., votes, release date, developer, author playtime) were excluded. Sentiment scores were generated for each review using a multilingual BERT-based model, producing a labeled dataset suitable for both sentiment and topic analysis. This collection provides a large-scale, balanced foundation for examining how price interact to shape review sentiment.

TABLE I. DATASET SUMMARY TABLE

Attribute	Value/Description	
Total Games	743	
Total Reviews	730,945	
Average Review per Game	≈ 1,000	
Price Segments	Low	1-20 USD
	Mid	21-40 USD
	High	>40 USD
Additional Column	game_id, votes_up, votes_down, release_date, author_playtime	

IV. RESULT & DISCUSSION

The Steam dataset was used with a pretrained mBERT model to identify the sentiment of each review. The dataset was then divided into three segments based on price. These segments were analyzed to explore the correlation between price and user sentiment, as well as for topic analysis.

A. Sentiment by Price Segment

We first examined the average numeric sentiment across price segments (Fig. 5). Low-priced games exhibit slightly higher average sentiment compared to mid- and high-priced games (High: 3.201, Mid: 3.341, Low: 3.412). Specifically, the sentiment for mid-priced games is about 4.3% higher than high-priced games, and for low-priced games it is about 6.6% higher than high-priced games.

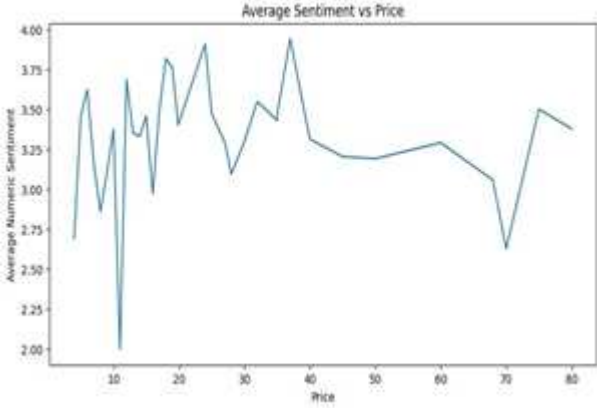


Fig. 5. Average sentiment per price segment.

This suggests that lower-priced games tend to receive slightly better player reception compared to higher-priced games.

B. Regression Analysis

To assess the relationship between price and sentiment, we performed Ordinary Least Squares[13] (OLS) regressions(Fig. 4) Using the following formula,

$$\text{numeric_sentiment} = \beta_0 + \beta_1 * \text{price}$$

where numeric_sentiment refers to the numerical sentiment values collected via M-BERT.

Model:

- price $\rightarrow$ numeric_sentiment
- $\beta = -0.0045$, $p < 0.001$, $R^2 = 0.002$

TABLE II. OLS REGRESSION RESULT FOR PRICE AND SENTIMENT CORRELATION

Variable	Coefficient(β)	Std. Error	t-value	p-value
Intercept(β_0)	3.4798	0.004	943.53	<0.001
Price(β_1)	-0.0045	0.000	-38.30	<0.001

Interpretation:

- For every 1 unit increase in price, the numerical sentiment score decreases by 0.0045 on average.
- Overall, the sentiment of reviews tends to decrease with increasing price, although price explains only a small fraction of the variation in sentiment ($R^2 = 0.002$)

C. Keyword-based Topic Analysis

We next analyzed 730,945 review contents using a manually curated keyword dictionary to categorize topics such as Story, Gameplay, Graphics, Value/Money, and others (Fig. 6, Fig. 7, Fig. 8). Across these reviews, the most frequently discussed 3 topics were:

1. Gameplay (120,355 counts)
2. Story (119,559)
3. Value for money (72,386)

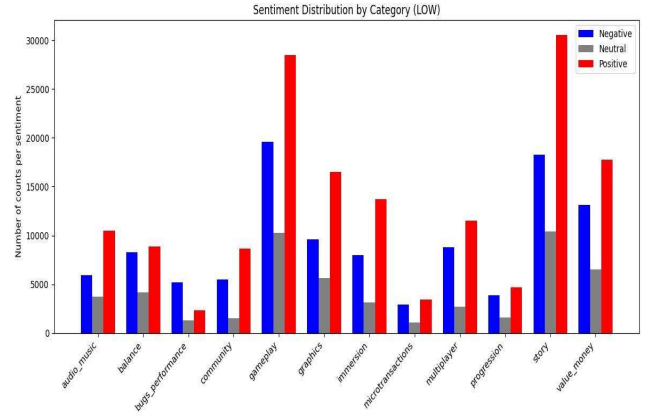


Fig. 6. Sentiment counts in different price segments in low priced games

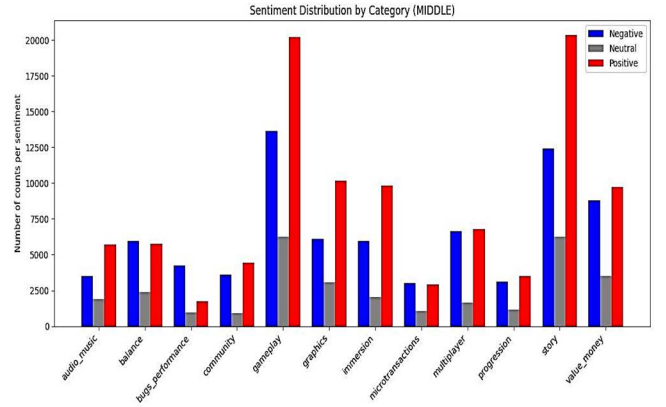


Fig. 7. Sentiment counts in different price segments in mid priced games

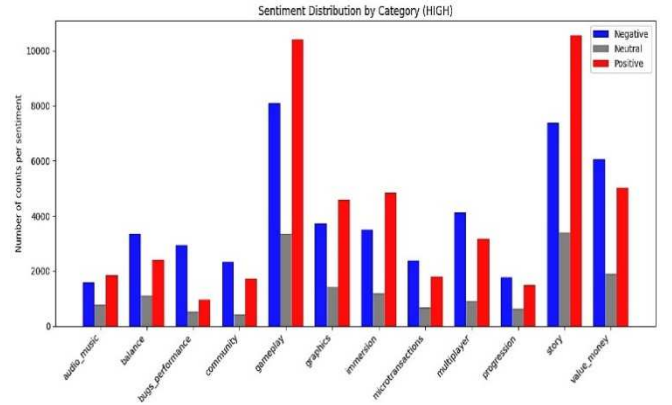


Fig. 8. Sentiment counts in different price segments in high priced games

For all price groups, gameplay and story consistently receive the highest volume of reviews, indicating they are the most discussed aspects. Positive sentiment dominates these categories across the board. However, bugs & performance and multiplayer are consistent sources of negative feedback, particularly among lower-priced products. Neutral reviews are generally a smaller portion of the total, but are still present in all categories. This suggests that while there is mixed

sentiment, a clear positive or negative opinion is more common in the reviews.

The findings of this study suggest that a game's price, while not a strong predictor of overall review sentiment, acts as a significant factor in shaping player expectations and the topics they choose to discuss. The research confirms that the correlation between price and sentiment is negligible, explaining only about 0.2% of the variation. While a slight trend indicates that lower-priced games receive slightly more positive sentiment on average, this correlation is weak and not a determining factor for a review's positivity or negativity.

The most impactful finding is the clear shift in review content across different price tiers, directly addressing the second research question. Low-priced games elicit detailed reviews that heavily emphasize value for money and story. In contrast, mid-priced games concentrate on gameplay and immersion. Finally, high-priced games generate fewer and narrower discussions, suggesting a more focused and possibly critical player base. This demonstrates that price serves as a "signal" that frames how players evaluate their experience.

Despite the differences in focus, certain trends persist across all price groups. Positive sentiment consistently dominates the gameplay and story categories, which are the most discussed aspects of a game. Conversely, bugs & performance and multiplayer features are common sources of negative feedback, particularly for lower-priced products.

V. CONCLUSION

In conclusion, this study has successfully addressed the research gap concerning the relationship between game pricing, review sentiment, and the content of player discourse. We found that a game's price does not significantly affect overall sentiment. Instead, it influences the specific topics players choose to discuss and evaluate. The distinct shifts in focus from "value" and "story" in low-priced games to "gameplay" and "immersion" in mid-priced games highlight the nuanced role of price as a framework for player expectations.

This research contributes to the field of game analytics by providing a deeper understanding of how pricing shapes the structure of player feedback, beyond simple sentiment prediction. For game developers, these findings offer valuable insights into the expectations of different player segments based on price point. Future research could expand upon this by incorporating other metadata, such as release date, developer, and publisher information, to provide a more comprehensive analysis of the factors that shape player reviews.

This work could be expanded in future research by testing generalizability by reproducing the analysis on different platforms (such as the Xbox, PlayStation, and mobile stores). Further insights might be obtained by treating price more precisely and adding more metadata (such as release date, genre, and developer reputation). Additionally, using unsupervised topic modeling techniques like LDA could reveal new themes outside of pre-made keyword dictionaries.

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